TOOL:R Programming **RegisterNo:192424335**

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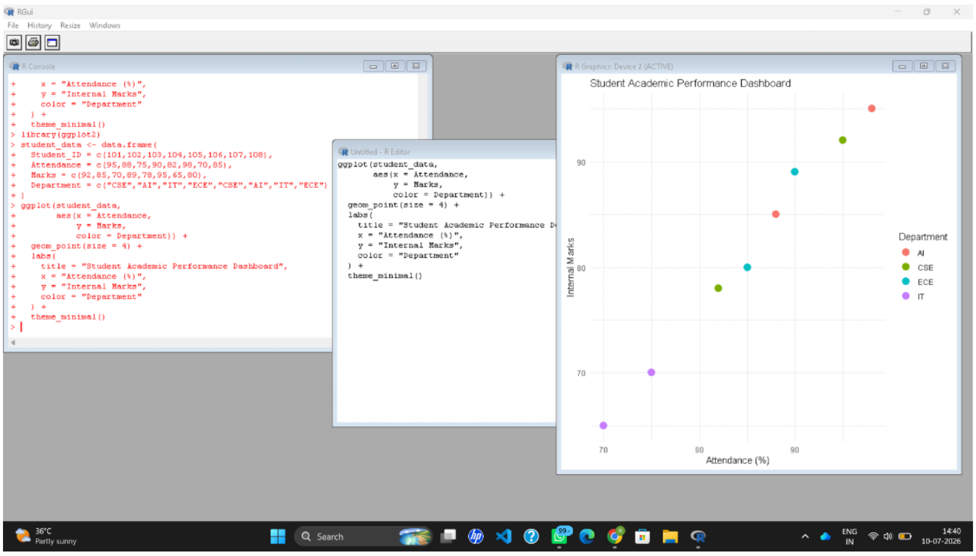
COURSE:Data Handling and Visualization **Code: DSA0604**

1.Student Academic performance Dashboard

Code:

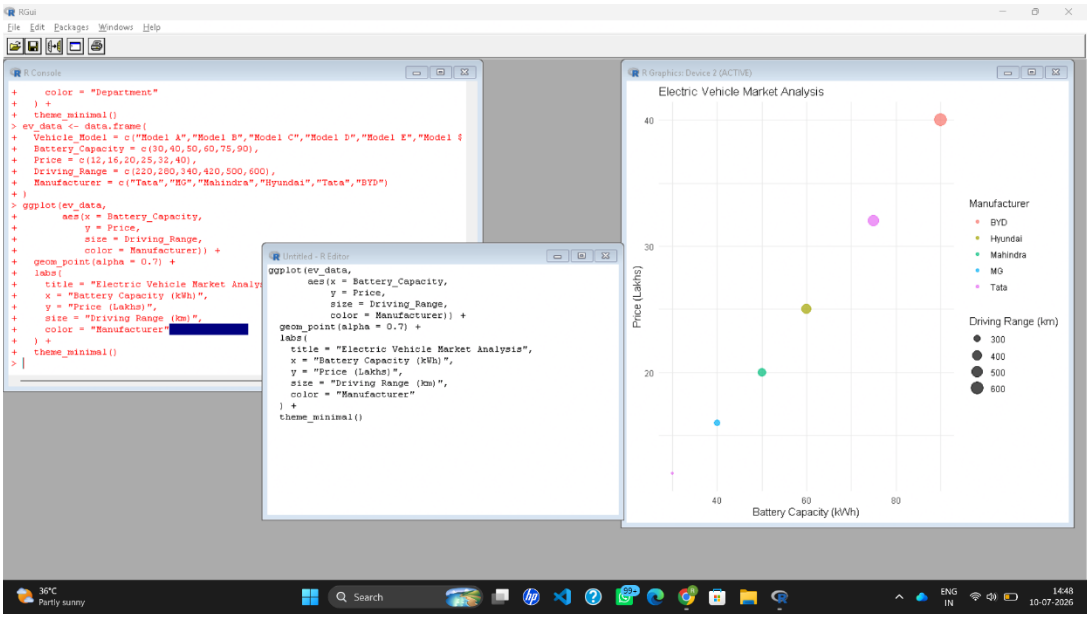
```
library(ggplot2)
data <- data.frame(
  Attendance = c(85,90,75,60,95,80),
  Marks = c(80,90,70,55,95,78),
  Department = c("CSE","AI","IT","ECE","CSE","AI")
)
ggplot(data, aes(x=Attendance, y=Marks, color=Department)) +
  geom_point(size=4) +
  labs(title="Student Academic Performance",
       x="Attendance (%)",
       y="Internal Marks")
```



Scenario 2: Electric Vehicle Market Analysis

Code:

```
library(ggplot2)

data <- data.frame(

  Battery=c(30,40,50,60,70),

  Price=c(8,10,14,18,22),

  Range=c(200,250,300,350,400),

  Manufacturer=c("A","B","A","C","B")

)

ggplot(data, aes(Battery, Price,

            size=Range,

            color=Manufacturer)) +

  geom_point() +

  labs(title="EV Market Analysis")
```

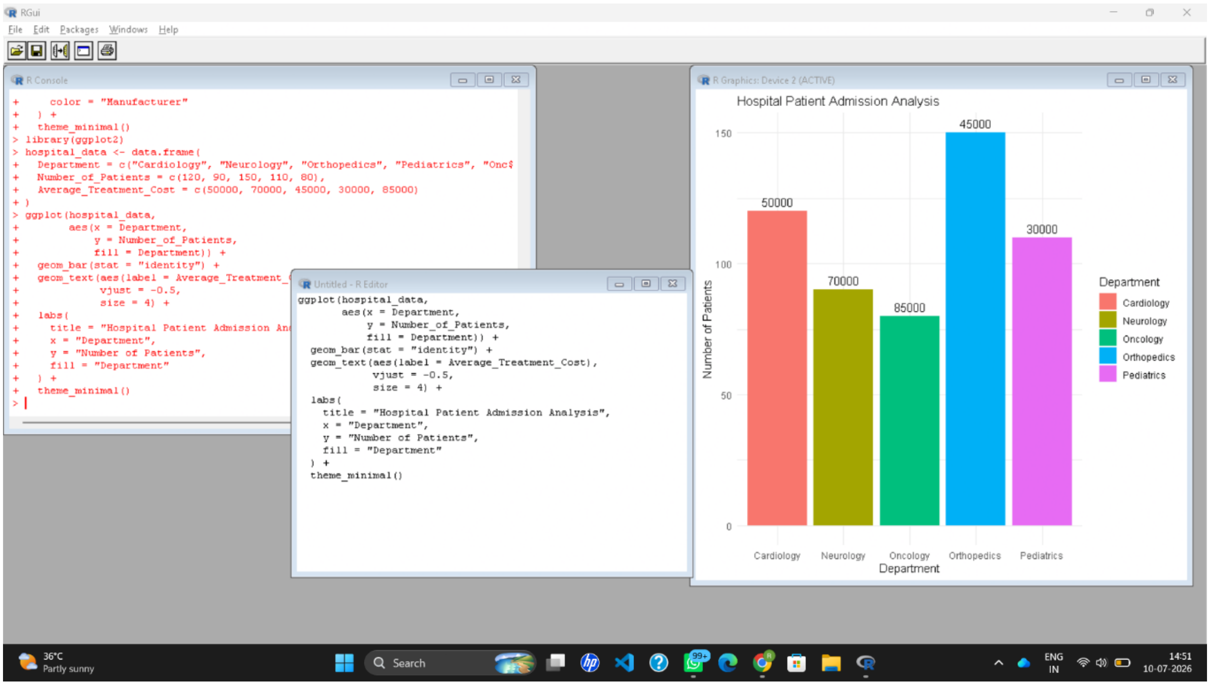


Scenario 3: Hospital Patient Admission Analysis

Code:

```
library(ggplot2)

data <- data.frame(

  Department=c("Cardiology","ENT","Ortho","Neuro"),

  Patients=c(120,90,150,110),

  Cost=c(5000,3000,7000,6500)
```

```
)
ggplot(data,
      aes(x=Department,
          y=Patients,
          fill=Department)) +
  geom_bar(stat="identity") +
  geom_text(aes(label=Cost),
          vjust=-0.5) +
  labs(title="Hospital Patient Admissions")
```

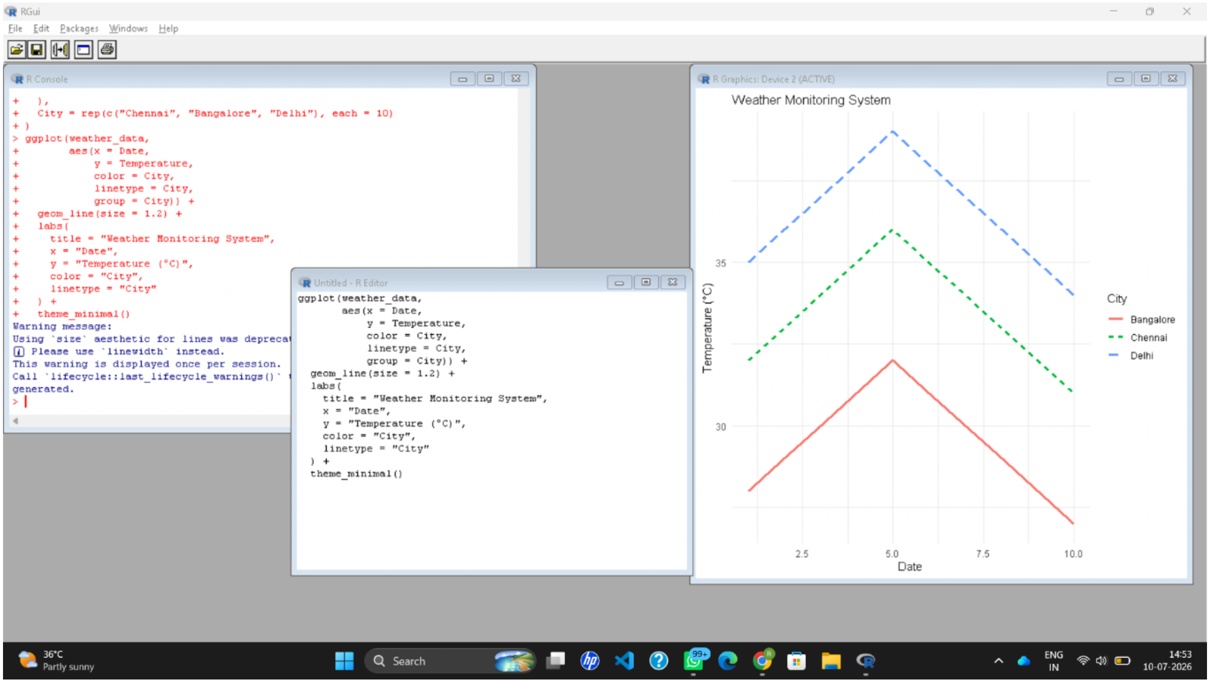


Scenario 4: Weather Monitoring System

Code:

```
library(ggplot2)
data <- data.frame(
  Date=1:6,
  Temperature=c(30,31,32,29,33,34,
              28,29,30,31,32,33,
              26,27,28,29,30,31),
  City=rep(c("Chennai","Delhi","Bangalore"),each=6)
```

```
)
ggplot(data,
      aes(Date,Temperature,
         color=City,
         linetype=City)) +
  geom_line(size=1) +
  labs(title="Weather Monitoring")
```

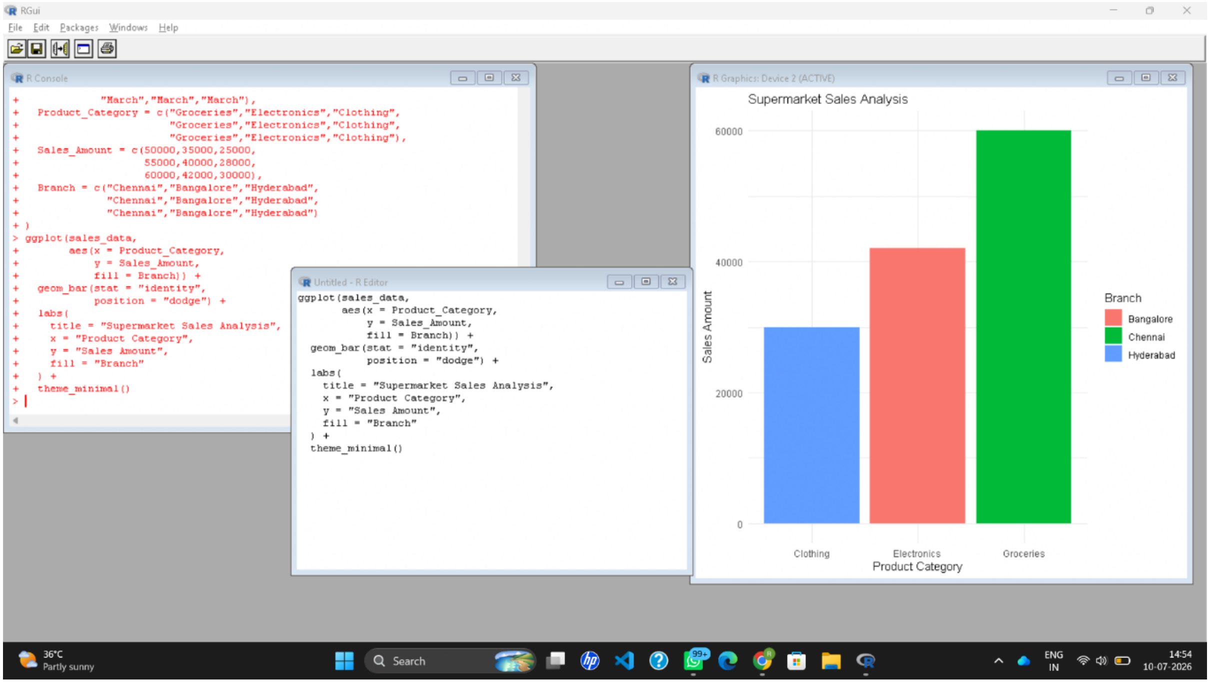


Scenario 5: Supermarket Sales Analysis

Code:

```
library(ggplot2)
data <- data.frame(
  Category=c("Food","Food","Drinks","Drinks"),
  Sales=c(100,120,80,95),
  Branch=c("A","B","A","B")
)
ggplot(data,
      aes(Category,Sales,
         fill=Branch)) +
```

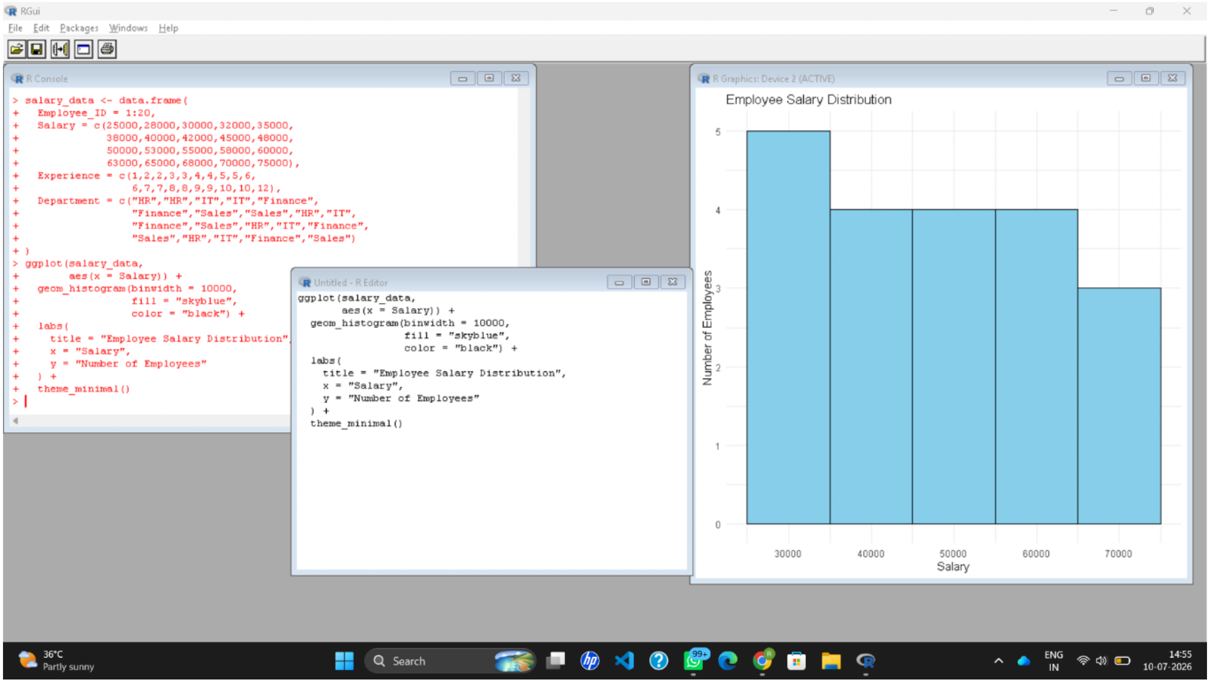
```
 geom_bar(stat="identity",
          position="dodge") +
 labs(title="Supermarket Sales")
```



Scenario 6: Employee Salary Distribution

Code:

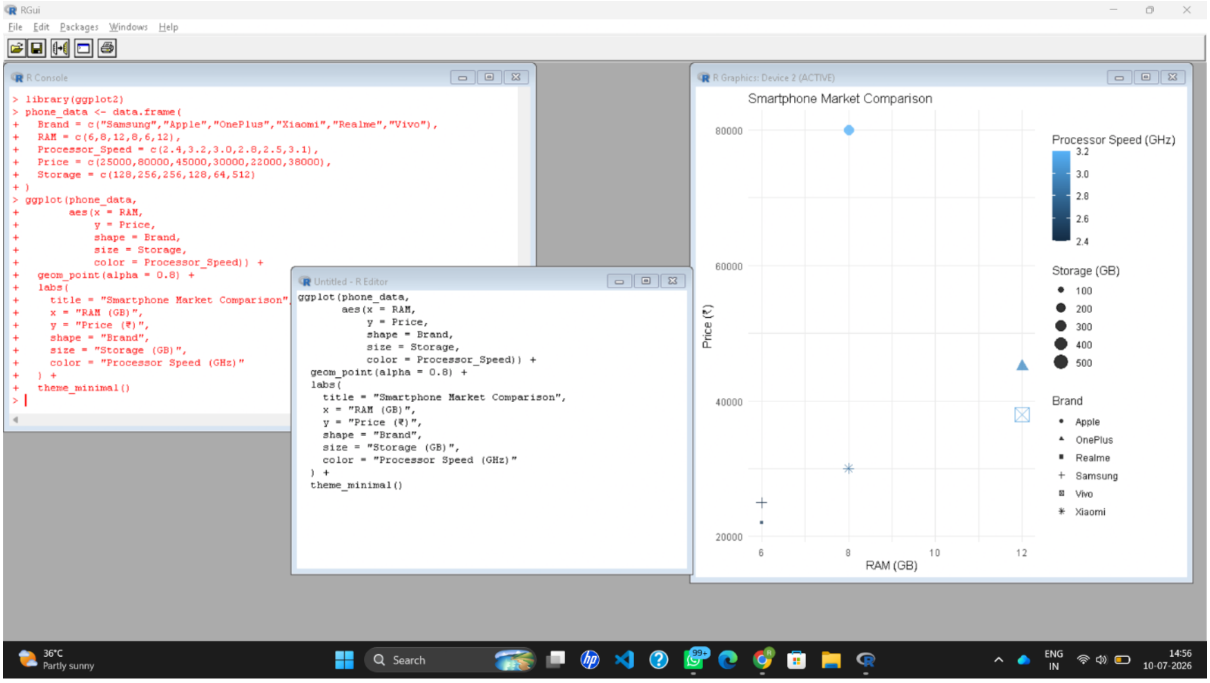
```
library(ggplot2)
data <- data.frame(
 Salary=c(25000,28000,30000,35000,40000,
          45000,50000,55000,60000,70000)
)
ggplot(data,
      aes(Salary)) +
 geom_histogram(binwidth=5000,
                fill="skyblue",
                color="black") +
 labs(title="Salary Distribution")
```



Scenario 7: Smartphone Market Comparison

Code:

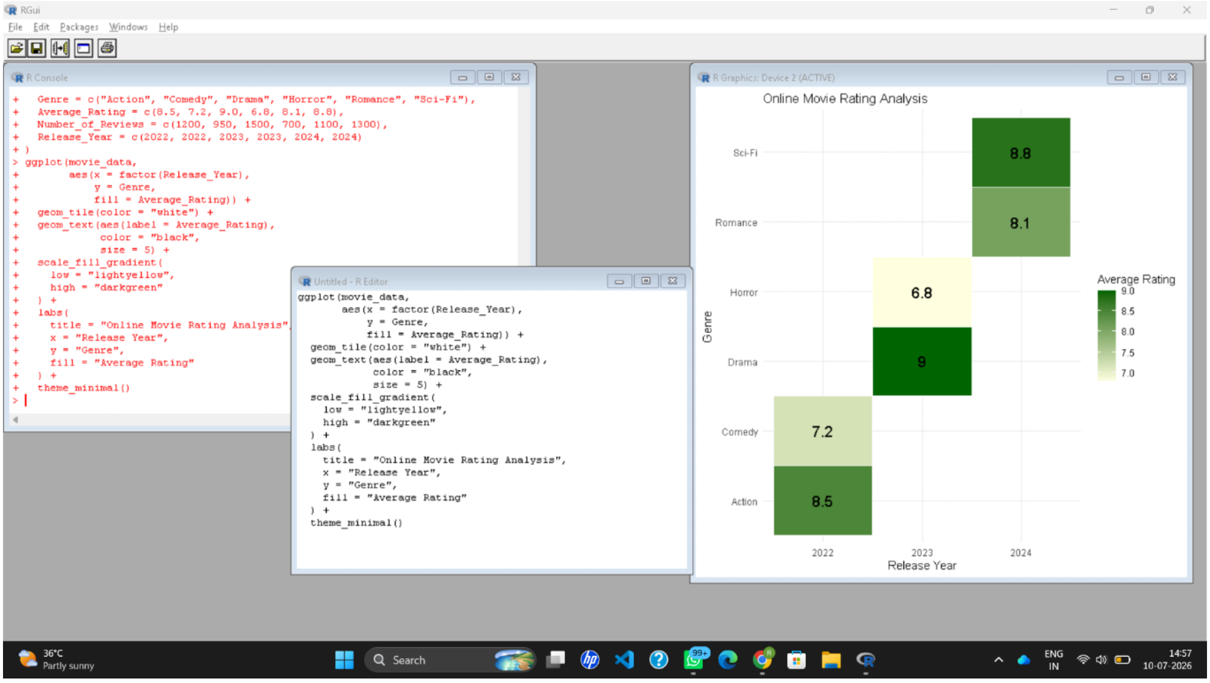
```
library(ggplot2)

data <- data.frame(
  RAM=c(4,6,8,12,16),
  Price=c(15000,22000,28000,45000,60000),
  Brand=c("Samsung","Vivo","Realme","Apple","OnePlus"),
  Storage=c(64,128,128,256,512),
  Speed=c(2.0,2.2,2.5,3.0,3.2)
)

ggplot(data,
       aes(RAM,Price,
           shape=Brand,
           size=Storage,
           color=Speed)) +
  geom_point() +
  labs(title="Smartphone Market")
```



Scenario 8: Online Movie Rating Analysis

Code:

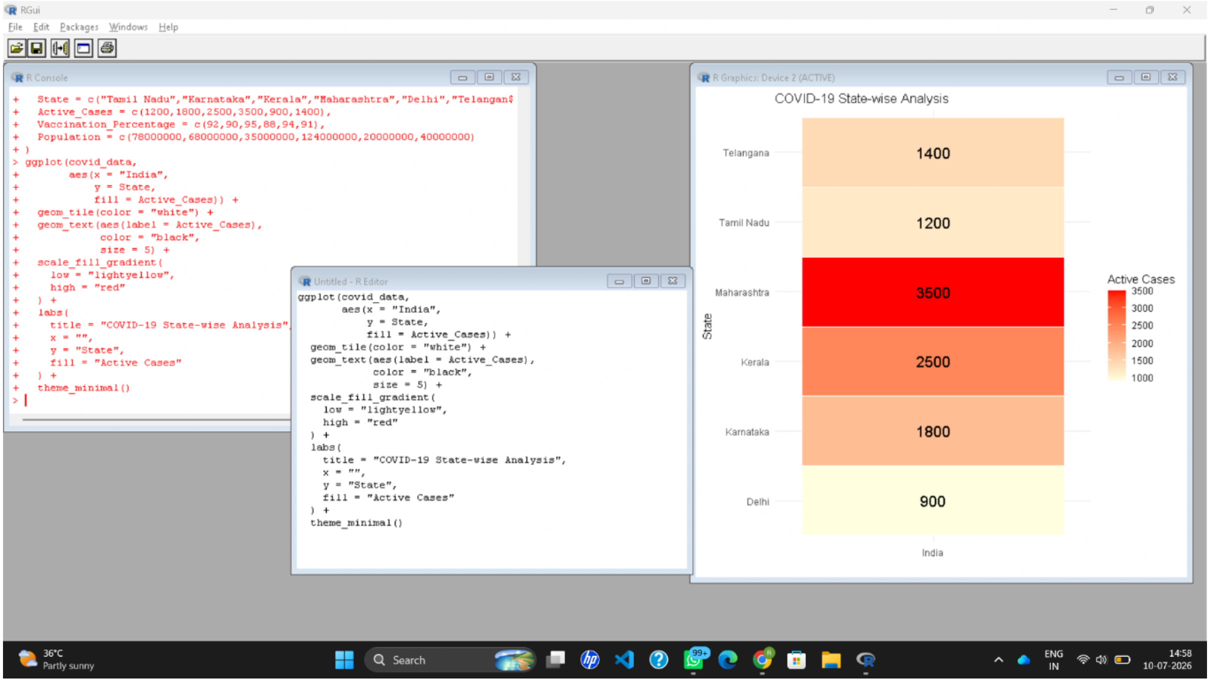
```
library(ggplot2)

data <- data.frame(

  Genre=c("Action","Comedy","Drama","Horror"),

  Year=c(2022,2022,2022,2022),

  Rating=c(4.5,3.8,4.8,3.5)

)


ggplot(data,

       aes(Genre,

           factor(Year),

           fill=Rating)) +

  geom_tile() +

  scale_fill_gradient(low="yellow",

                      high="red") +

  labs(title="Movie Rating Heatmap")
```



Scenario 9: COVID-19 State-wise Analysis

Code:

```
library(ggplot2)

data <- data.frame(

  State=c("Tamil Nadu","Kerala","Karnataka","Andhra"),

  Cases=c(2000,3500,2800,1500)

)

ggplot(data,

      aes(State,"COVID",

          fill=Cases)) +

  geom_tile() +

  scale_fill_gradient(low="lightblue",

                      high="darkred") +

  labs(title="COVID State Analysis")
```



Scenario 10: Iris Flower Classification

Code:

```
library(ggplot2)

data(iris)

ggplot(iris,

    aes(Sepal.Length,

        Petal.Length,

        color=Species,

        shape=Species,

        size=Sepal.Width)) +

  geom_point() +

  labs(title="Iris Flower Classification")
```


RGui
File History Resize Windows
R Console
> head(iris)
Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1 5.1 3.5 1.4 0.2 setosa
2 4.9 3.0 1.4 0.2 setosa
3 4.7 3.2 1.3 0.2 setosa
4 4.6 3.1 1.5 0.2 setosa
5 5.0 3.6 1.4
6 5.4 3.9 1.7
> ggplot(iris,
+ aes(x = Sepal.Length,
+ y = Petal.Length,
+ color = Species,
+ shape = Species,
+ size = Sepal.Width)) +
+ geom_point(alpha = 0.8) +
+ labs(
+ title = "Iris Flower Classification",
+ x = "Sepal Length (cm)",
+ y = "Petal Length (cm)",
+ color = "Species",
+ shape = "Species",
+ size = "Sepal Width (cm)"
+) +
+ theme_minimal()
Untitled - R Editor
ggplot(iris,
aes(x = Sepal.Length,
y = Petal.Length,
color = Species,
shape = Species,
size = Sepal.Width)) +
geom_point(alpha = 0.8) +
labs(
title = "Iris Flower Classification",
x = "Sepal Length (cm)",
y = "Petal Length (cm)",
color = "Species",
shape = "Species",
size = "Sepal Width (cm)"
) +
theme_minimal()
R Graphics: Device 2 (ACTIVE)
Iris Flower Classification
Petal Length (cm)
6
4
2
5
6
7
8
Sepal Length (cm)
Sepal Width (cm)
2.0
2.5
3.0
3.5
4.0
Species
setosa
versicolor
virginica
35°C
Partly sunny
Search
ENG
IN
17:40
10-07-2026